

## DM54LS491/74LS491 10-Bit Counter

### General Description

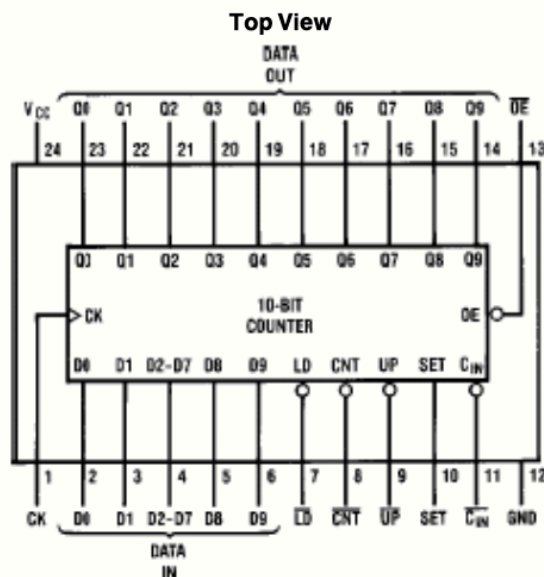
The ten-bit counter can count up, count down, set, and load 2 LSB's, 2 MSB's and 6 middle bits high or low as a group. All operations are synchronous with the clock. SET overrides LOAD, COUNT and HOLD. LOAD overrides COUNT. COUNT is conditional on  $C_{IN}$ , otherwise it holds.

All outputs are enabled when  $\overline{OE}$  is low, otherwise HIGH-Z. The 24 mA  $I_{OL}$  outputs are suitable for driving RAM/PROM address lines in video graphics systems.

### Features/Benefits

- CRT vertical and horizontal timing generation
- Bus-structured pinout
- 24-pin SKINNYDIP saves space
- TRI-STATE® outputs drive bus lines
- Low current PNP inputs reduce loading

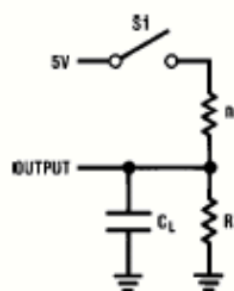
### Connection Diagram



TL/L/8332-1

Order Number DM54LS491J,  
DM74LS491J or DM74LS491N  
See NS Package Number J24F or N24C

### Standard Test Load



TL/L/8332-2

### Function Table

| OE | CK | SET | LD | CNT | $C_{IN}$ | UP | D9-D0 | Q9-Q0     | Operation    |
|----|----|-----|----|-----|----------|----|-------|-----------|--------------|
| H  | X  | X   | X  | X   | X        | X  | X     | Z         | Hi-Z         |
| L  | ↑  | H   | X  | X   | X        | X  | X     | H         | Set all HIGH |
| L  | ↑  | L   | L  | X   | X        | X  | D     | D         | LOAD D       |
| L  | ↑  | L   | H  | H   | X        | X  | X     | Q         | HOLD         |
| L  | ↑  | L   | H  | L   | H        | X  | X     | Q         | HOLD         |
| L  | ↑  | L   | H  | L   | L        | L  | X     | Q plus 1  | Count UP     |
| L  | ↑  | L   | H  | L   | L        | H  | X     | Q minus 1 | Count DN     |

## Absolute Maximum Ratings

If Military/Aerospace specified devices are required, please contact the National Semiconductor Sales Office/Distributors for availability and specifications.

Supply Voltage  $V_{CC}$  7V  
Input Voltage 5.5V

Off-State Output Voltage  
Storage Temperature

5.5V  
−65° to +150°C

## Operating Conditions

| Symbol   | Parameter                      | Military |     |      | Commercial |     |      | Units |
|----------|--------------------------------|----------|-----|------|------------|-----|------|-------|
|          |                                | Min      | Typ | Max  | Min        | Typ | Max  |       |
| $V_{CC}$ | Supply Voltage                 | 4.5      | 5   | 5.5  | 4.75       | 5   | 5.25 | V     |
| $T_A$    | Operating Free-Air Temperature | −55      |     | 125* | 0          |     | 75   | °C    |
| $t_w$    | Width of Clock                 | High     | 40  |      | 40         |     |      | ns    |
|          |                                | Low      | 35  |      | 35         |     |      |       |
| $t_{SU}$ | Set-Up Time                    | 60       |     |      | 50         |     |      | ns    |
| $t_h$    | Hold Time                      | 0        | −15 |      | 0          | −15 |      |       |

\* Case temperature

## Electrical Characteristics Over Operating Conditions

| Symbol    | Parameter                     | Test Conditions                                                             |                               | Min | Typ† | Max   | Units |
|-----------|-------------------------------|-----------------------------------------------------------------------------|-------------------------------|-----|------|-------|-------|
| $V_{IL}$  | Low-Level Input Voltage       |                                                                             |                               |     |      | 0.8   | V     |
| $V_{IH}$  | High-Level Input Voltage      |                                                                             |                               | 2   |      |       | V     |
| $V_{IC}$  | Input Clamp Voltage           | $V_{CC} = \text{MIN}$                                                       | $I_I = -18 \text{ mA}$        |     |      | −1.5  | V     |
| $I_{IL}$  | Low-Level Input Current       | $V_{CC} = \text{MAX}$                                                       | $V_I = 0.4 \text{ V}$         |     |      | −0.25 | mA    |
| $I_{IH}$  | High-Level Input Current      | $V_{CC} = \text{MAX}$                                                       | $V_I = 2.4 \text{ V}$         |     |      | 25    | μA    |
| $I_I$     | Maximum Input Current         | $V_{CC} = \text{MAX}$                                                       | $V_I = 5.5 \text{ V}$         |     |      | 1     | mA    |
| $V_{OL}$  | Low-Level Output Voltage      | $V_{CC} = \text{MIN}$<br>$V_{IL} = 0.8 \text{ V}$<br>$V_{IH} = 2 \text{ V}$ | MIL $I_{OL} = 12 \text{ mA}$  | 2.4 |      | 0.5   | V     |
|           |                               |                                                                             | COM $I_{OL} = 24 \text{ mA}$  |     |      |       |       |
| $V_{OH}$  | High-Level Output Voltage     | $V_{CC} = \text{MIN}$<br>$V_{IL} = 0.8 \text{ V}$<br>$V_{IH} = 2 \text{ V}$ | MIL $I_{OH} = -2 \text{ mA}$  | 2.4 |      |       | V     |
|           |                               |                                                                             | COM $I_{OH} = 3.2 \text{ mA}$ |     |      |       |       |
| $I_{OZL}$ | Off-State Output Current      | $V_{CC} = \text{MAX}$<br>$V_{IL} = 0.8 \text{ V}$<br>$V_{IH} = 2 \text{ V}$ | $V_O = 0.4 \text{ V}$         |     |      | −100  | μA    |
| $I_{OZH}$ |                               |                                                                             | $V_O = 2.4 \text{ V}$         |     |      | 100   | μA    |
| $I_{OS}$  | Output Short-Circuit Current* | $V_{CC} = 5.0 \text{ V}$                                                    | $V_O = 0 \text{ V}$           | −30 |      | −130  | mA    |
| $I_{CC}$  | Supply Current                | $V_{CC} = \text{MAX}$                                                       |                               |     | 120  | 180   | mA    |

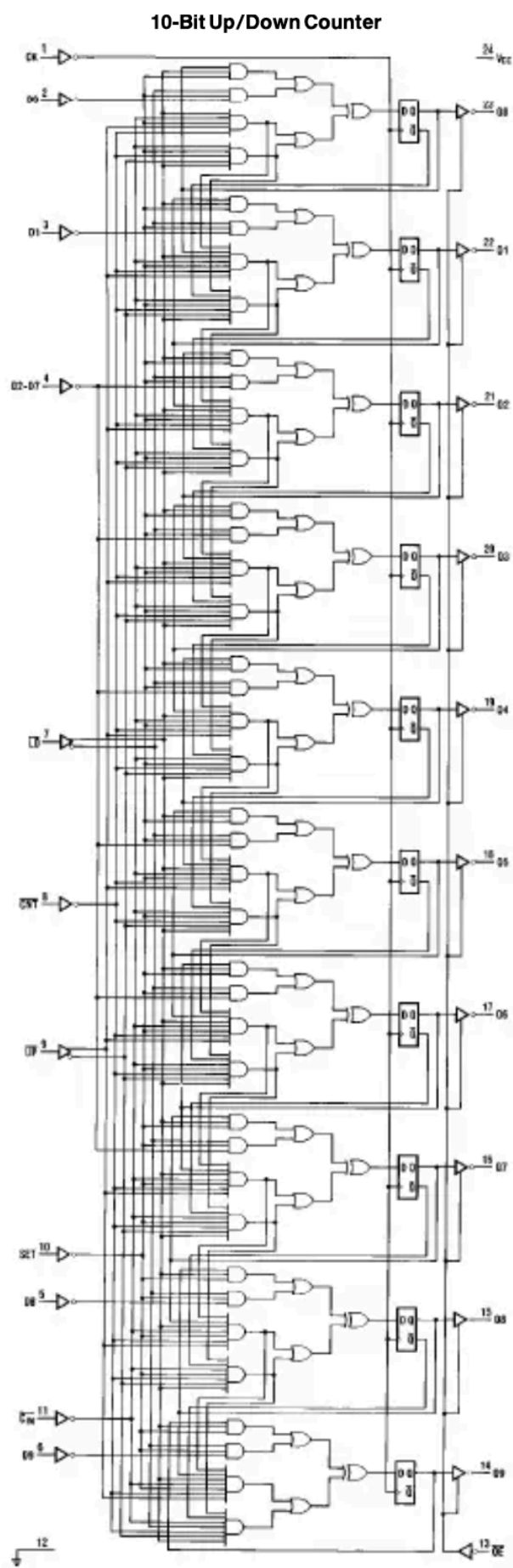
\* No more than one output should be shorted at a time and duration of the short-circuit should not exceed one second.

† All typical values are at  $V_{CC} = 5 \text{ V}$ ,  $T_A = 25^\circ \text{C}$

## Switching Characteristics Over Operating Conditions

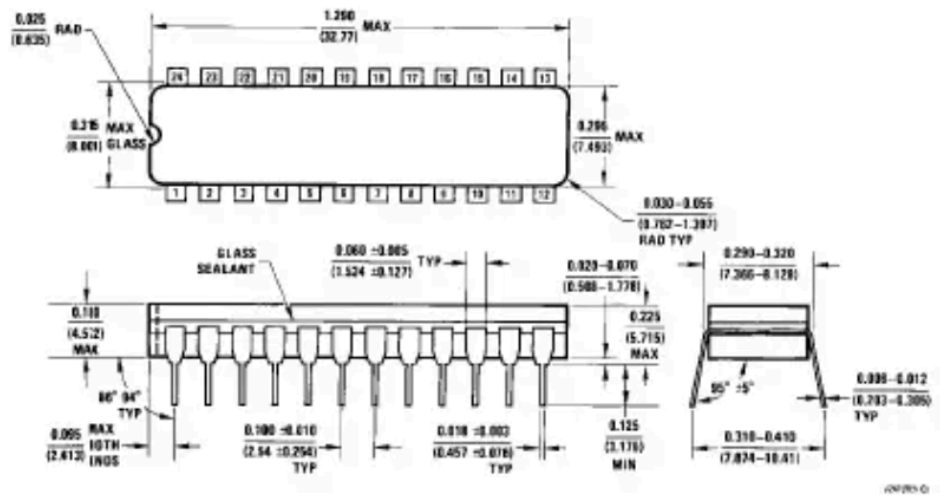
| Symbol    | Parameter               | Test Conditions<br>(See Test Load)                                | Military |     |     | Commercial |     |     | Units |
|-----------|-------------------------|-------------------------------------------------------------------|----------|-----|-----|------------|-----|-----|-------|
|           |                         |                                                                   | Min      | Typ | Max | Min        | Typ | Max |       |
| $f_{MAX}$ | Maximum Clock Frequency | $C_L = 50 \text{ pF}$<br>$R_1 = 200 \Omega$<br>$R_2 = 390 \Omega$ | 10.5     |     |     | 12.5       |     |     | MHz   |
| $t_{PD}$  | Clock to Q              |                                                                   |          | 20  | 35  |            | 20  | 30  | ns    |
| $t_{PZX}$ | Output Enable Delay     |                                                                   |          | 35  | 55  |            | 35  | 45  | ns    |
| $t_{PXZ}$ | Output Disable Delay    |                                                                   |          | 35  | 55  |            | 35  | 45  | ns    |

# Logic Diagram

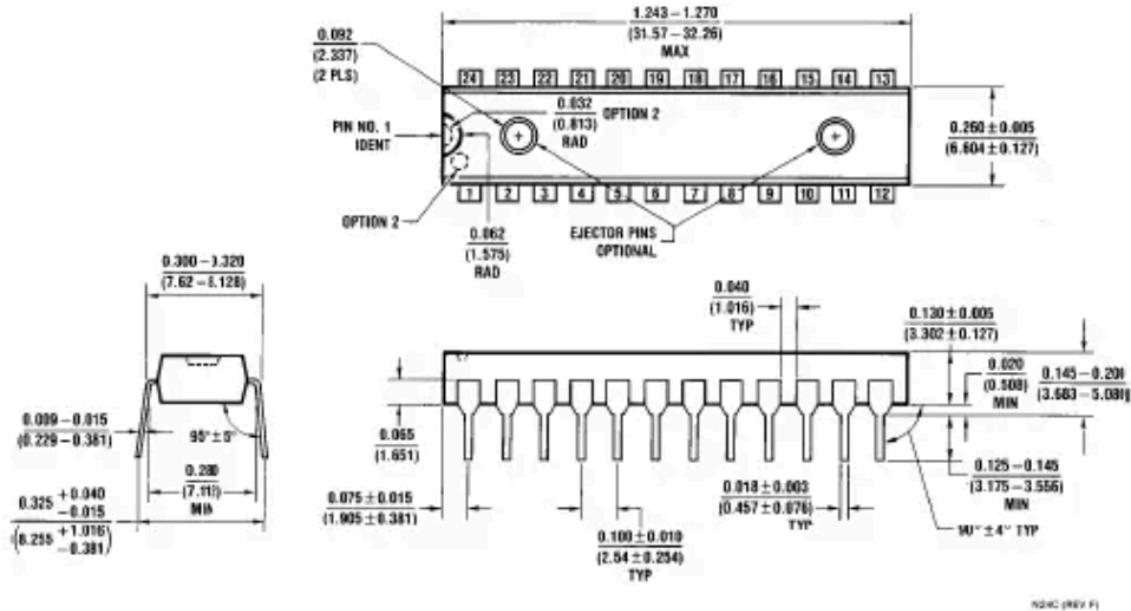


TL/L/8332-3

# Physical Dimensions inches (millimeters)



**24-Pin Narrow Ceramic Dual-In-Line Package (J)**  
**Order Number DM54LS491J or DM74LS491J**  
**NS Package Number J24F**



**24-Pin Narrow Plastic Dual-In-Line Package (N)**  
**Order Number DM74LS491N**  
**NS Package Number N24C**

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